THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 3: **The World of Metals and Non-metals** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A substance is shiny, can be hammered into thin sheets and conducts electricity. It is probably a: 1

(a) non-metal **(b)** gas **(c)** liquid non-metal **(d)** metal

2. When magnesium ribbon is burnt and the ash dissolved in water, the solution turns: 1

(a) colourless **(b)** red litmus blue **(c)** green **(d)** blue litmus red

3. Why does an aluminium vessel not corrode easily? 1

(a) it is sonorous **(b)** a protective oxide layer forms
(c) it is brittle **(d)** it is a non-metal

4. A substance is dull, brittle and does not conduct electricity. It is probably a: 1

(a) metal **(b)** non-metal **(c)** magnet **(d)** alloy

5. When sulphur is burnt and gas passed into water, the solution turns: 1

(a) red litmus blue **(b)** blue litmus red **(c)** green **(d)** no change

6. Why is silver used in mirrors and in jewellery? 1

(a) it is brittle **(b)** it is lustrous **(c)** it is gaseous **(d)** it is dull

7. Which of these is a metal that is also a poor conductor of heat compared with others? 1

(a) silver **(b)** lead **(c)** copper **(d)** aluminium

8. When carbon dioxide is passed through lime water it turns: 1

(a) red **(b)** milky **(c)** green **(d)** blue

9. Why are electric cables made of copper wires coated with PVC? 1

(a) copper conducts, PVC insulates **(b)** PVC is a conductor
(c) copper is an insulator **(d)** both are metals

10. Which of these is a non-metal that conducts electricity? 1

(a) phosphorus **(b)** chlorine **(c)** graphite **(d)** sulphur

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Why do silver articles turn black on keeping in air? 2

12. Give two examples where metal oxides are used as bases. 2

13. Why do we store phosphorus under water? 2

14. Why does aluminium not rust though it is reactive? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Predict what will happen if a copper nail is dipped in iron sulphate solution and explain. 3

16. Explain why a tarnished copper vessel can be cleaned with lemon or tamarind. 3

17. How do metals help in conduction of heat in cooking? 3

OR

Describe the activity to compare the reactivity of zinc, iron and copper with dilute HCl. 3

18. Compare the properties of iron and sulphur. 3

OR

Why are chemical formulae written as MgO and SO_2? Write the names. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Plan an investigation to classify ten given materials as metals or non-metals using lustre, hardness, conduction and sound. 5

OR

Describe a safe procedure to burn magnesium and test the product. 5

20. Explain the formation of oxides of magnesium and sulphur, their tests and equations. 5

OR

Explain the importance of metals and non-metals to living beings and industry. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND